1. Project Overview

This project analyses e-commerce sales data to understand performance across product categories, months, customer cities and payment methods. Microsoft Excel was used for exploratory analysis and an interactive dashboard. Microsoft Power BI was used for data modelling, DAX measures and interactive reporting.

2. Project Objectives

• Prepare and review raw e-commerce transaction data.

• Summarise sales using Excel PivotTables and dashboard visuals.

• Create key performance indicators (KPIs) in Power BI using DAX.

• Analyse sales by category, city, month and payment method.

• Provide interactive filters for customer and sales segments.

• Present findings in a clear, business-friendly format.

3. Dataset Description

The dataset contains order, customer, product, date, quantity, pricing, sales, payment and customer-segment fields. Fields used include Order ID, Customer ID, Order Date, Product ID, Quantity, Payment Method, Payment Status, Customer Age, Customer City, Customer Segment, Product Name, Category, Unit Price, Sales and Order Value.

4. Data Cleaning and Preparation

• Loaded the source data into Excel and reviewed field names, data types and records.

• Prepared a clean-data workbook for analysis.

• Loaded the clean dataset into Power BI and checked fields used in visuals and measures.

• During Power BI validation, records with a null Category value were excluded from the category analysis to avoid an unwanted blank category.

• Checked date, category, city, payment and customer-segment fields.

Data-quality note: Because records with a null Category were filtered in Power BI, totals in Excel and Power BI may differ. Use the same filtered dataset when making direct comparisons.

5. Excel Analysis and Dashboard

• Created PivotTables for category-wise sales, payment-method sales, city-wise sales and monthly sales trends.

• Built an Excel dashboard with charts for the four analysis areas.

• Added slicers for Customer City, Category, Payment Method and Customer Segment.

• Connected slicers to the relevant PivotTables so users can filter the dashboard.

• Created KPI summaries for sales, orders, customers, quantity and average order value.

6. Power BI Analysis and Dashboard

• Imported the clean dataset into Power BI.

• Created DAX measures for Total Sales, Total Orders, Total Customers, Total Quantity and Average Order Value.

• Used DISTINCTCOUNT for Order ID and Customer ID to calculate distinct order and customer counts.

• Created a monthly sales trend line chart, city-wise sales bar chart, category-wise sales column chart and payment-method donut chart.

• Added slicers for Customer Segment, Payment Method, Customer City and Category.

7. KPI Summary

The latest Power BI dashboard screenshot displays the following rounded KPI values:

KPI	Displayed value	Interpretation
Total Sales	3M (rounded)	Overall sales; exact value needs display units set to None.
Total Orders	49K (rounded)	Distinct order count displayed in thousands.
Total Quantity	93K (rounded)	Total units sold displayed in thousands.
Total Customers	9,820	Distinct customers.
Average Order Value	69.99	Average sales value per distinct order.

Note: 3M, 49K and 93K are rounded values visible in the supplied screenshot, not exact totals. Confirm exact values in Power BI before using them in a formal presentation.

8. Key Business Insights

• Electronics is the highest-sales category in the displayed category chart. Accessories is the next strongest category.

• Stationery and Gaming contribute comparatively lower sales than the leading categories.

• Mumbai is the leading city by sales in the displayed city chart, followed by Kolkata and Pune.

• Gateway contributes 48.39% of sales by payment method, followed by CardToCard at 24.11%, Wallet at 17.54% and Cash at 9.96%.

• The monthly sales line chart shows a general decline from January toward the later months, with a relatively stable pattern near year-end.

9. Conclusion

The project demonstrates an end-to-end analytics workflow: preparing data, summarising it in Excel, creating DAX measures in Power BI and presenting findings through interactive dashboards. The dashboards allow users to explore sales performance by category, city, month, payment method and customer segment. Further work could include profit and margin analysis, customer retention analysis, product-level performance and forecasting, if the required fields are available.

10. Project Deliverables

• Raw_Data.xlsx – original source dataset.

• Clean_Data.xlsx – prepared dataset used for analysis.

• Excel_Analysis.xlsx – Excel PivotTables and interactive dashboard.

• PowerBI_Dashboard.pbix – Power BI report and DAX measures.

• Project_Documentation.pdf – this report.

• Project_Demo.mp4 – screen recording explaining the analysis and dashboards.